

Администрирование локальных сетей

Лабораторная работа №15

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Цели и задачи работы

Настроить динамическую маршрутизацию между территориями организации.

Выполнение лабораторной работы

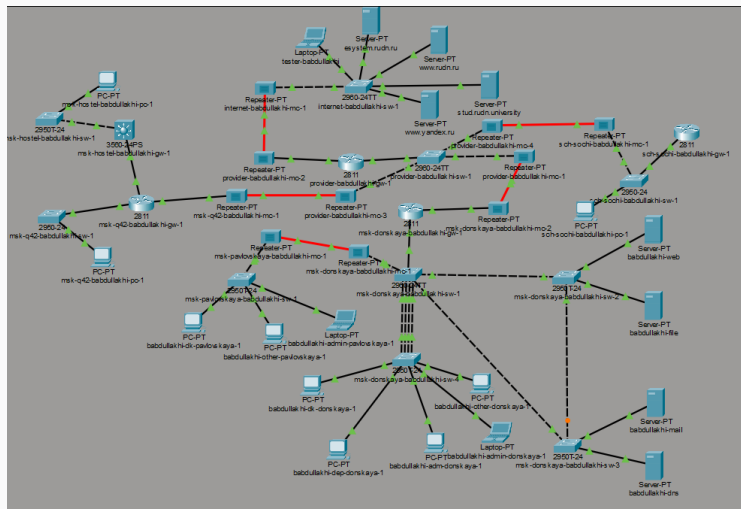


Рис. 1: Топология сети

Настройка OSPF на msk-donskaya-gw-1

```
msk-donskaya-babdullakhi-gw-1>conf t
^
% Invalid input detected at '^' marker.

msk-donskaya-babdullakhi-gw-1>en
Password:
msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#router ospf 1
msk-donskaya-babdullakhi-gw-1(config-router)#router-id 10.128.254.1
msk-donskaya-babdullakhi-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-donskaya-babdullakhi-gw-1(config-router)#exit
msk-donskaya-babdullakhi-gw-1(config)#exit
msk-donskaya-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-babdullakhi-gw-1#
```

Рис. 2: Настройка OSPF на маршрутизаторе msk-donskaya-gw-1

Настройка OSPF на msk-q42-gw-1

```
msk-q42-babdullakhi-gw-1>en
Password:
msk-q42-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-babdullakhi-gw-1(config)#router ospf 1
msk-q42-babdullakhi-gw-1(config-router)#router-id 10.128.254.3
msk-q42-babdullakhi-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-q42-babdullakhi-gw-1(config-router)#exit
msk-q42-babdullakhi-gw-1(config)#exit
msk-q42-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
msk-q42-babdullakhi-gw-1#
00:54:48: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/1.5 from LOADING to FULL,
Loading Done
msk-q42-babdullakhi-gw-1#
```

Рис. 3: Настройка OSPF на маршрутизаторе msk-q42-gw-1

Настройка OSPF на msk-hostel-gw-1

```
msk-hostel-babdullakhi-gw-1>en
Password:
msk-hostel-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-babdullakhi-gw-1(config)#router ospf 1
msk-hostel-babdullakhi-gw-1(config-router)#router-id 10.128.254.3
msk-hostel-babdullakhi-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
msk-hostel-babdullakhi-gw-1(config-router)#exit
msk-hostel-babdullakhi-gw-1(config)#exit
msk-hostel-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
msk-hostel-babdullakhi-gw-1#
00:59:47: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on Vlan202 from LOADING to FULL, Loading Done
msk-hostel-babdullakhi-gw-1#
```

Рис. 4: Настройка OSPF на маршрутизаторе msk-hostel-gw-1

Настройка OSPF на sch-sochi-gw-1

```
sch-sochi-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-babdullakhi-gw-1(config)#router ospf 1
sch-sochi-babdullakhi-gw-1(config-router)#router-id 10.128.254.4
sch-sochi-babdullakhi-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
sch-sochi-babdullakhi-gw-1(config-router)#exit
sch-sochi-babdullakhi-gw-1(config)#exit
sch-sochi-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-babdullakhi-gw-1#
00:57:17: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on FastEthernet0/0.6 from LOADING to FULL,
Loading Done

sch-sochi-babdullakhi-gw-1#
```

Рис. 5: Настройка OSPF на маршрутизаторе sch-sochi-gw-1

Создание VLAN 7 на коммутаторе провайдера

```
provider-babdullakhi-sw-1>en
Password:
provider-babdullakhi-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-babdullakhi-sw-1(config)#vlan 7
provider-babdullakhi-sw-1(config-vlan)#name q42-sochi
provider-babdullakhi-sw-1(config-vlan)#exit
provider-babdullakhi-sw-1(config)#int vlan 7
provider-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

provider-babdullakhi-sw-1(config-if)#no sh
provider-babdullakhi-sw-1(config-if)#exit
provider-babdullakhi-sw-1(config)#exit
provider-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
provider-babdullakhi-sw-1#
```

Рис. 6: Создание VLAN 7 на коммутаторе provider-sw-1

Настройка подинтерфейса на msk-q42-gw-1

```
msk-q42-babdullakhi-gw-1(config)#  
msk-q42-babdullakhi-gw-1(config)#int f0/1.7  
msk-q42-babdullakhi-gw-1(config-subif)#  
%LINK-3-UPDOWN: Interface FastEthernet0/1.7, changed state to down  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.7, changed state to up  
  
msk-q42-babdullakhi-gw-1(config-subif)#enc dot1q 7  
msk-q42-babdullakhi-gw-1(config-subif)#ip addr 10.128.255.9 255.255.255.252  
msk-q42-babdullakhi-gw-1(config-subif)#descr sochi  
msk-q42-babdullakhi-gw-1(config-subif)#exit  
msk-q42-babdullakhi-gw-1(config)#exit  
msk-q42-babdullakhi-gw-1#  
%SYS-5-CONFIG_I: Configured from console by console  
  
msk-q42-babdullakhi-gw-1#wr mem  
Building configuration...  
[OK]  
msk-q42-babdullakhi-gw-1#
```

Рис. 7: Настройка подинтерфейса FastEthernet0/1.7 на маршрутизаторе msk-q42-gw-1

Создание VLAN 7 на коммутаторе Сочи

```
sch-sochi-babdullakhi-sw-1>en
Password:
sch-sochi-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-babdullakhi-sw-1(config)#
sch-sochi-babdullakhi-sw-1(config)#vlan 7
sch-sochi-babdullakhi-sw-1(config-vlan)#name q42-sochi
sch-sochi-babdullakhi-sw-1(config-vlan)#int vlan 7
sch-sochi-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan7, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan7, changed state to up

sch-sochi-babdullakhi-sw-1(config-if)#no sh
sch-sochi-babdullakhi-sw-1(config-if)#exit
sch-sochi-babdullakhi-sw-1(config)#exit
sch-sochi-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
sch-sochi-babdullakhi-sw-1#
```

Рис. 8: Создание VLAN 7 на коммутаторе sch-sochi-sw-1

Настройка подинтерфейса на sch-sochi-gw-1

```
sch-sochi-babdullakhi-gw-1#
sch-sochi-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-babdullakhi-gw-1(config)#
sch-sochi-babdullakhi-gw-1(config)#int f0/0.7
sch-sochi-babdullakhi-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.7, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.7, changed state to up

sch-sochi-babdullakhi-gw-1(config-subif)#enc dot1q 7
sch-sochi-babdullakhi-gw-1(config-subif)#ip addr 10.128.255.10 255.255.255.252
sch-sochi-babdullakhi-gw-1(config-subif)#desc q42
sch-sochi-babdullakhi-gw-1(config-subif)#exit
sch-sochi-babdullakhi-gw-1(config)#exit
sch-sochi-babdullakhi-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-babdullakhi-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-babdullakhi-gw-1#
01:00:36: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.2 on FastEthernet0/0.7 from LOADING to FULL,
Loading Done

sch-sochi-babdullakhi-gw-1#
```

Рис. 9: Настройка подинтерфейса FastEthernet0/0.7 на маршрутизаторе sch-sochi-gw-1

Проверка процесса OSPF

```
msk-donskaya-babdullakhi-gw-1#
msk-donskaya-babdullakhi-gw-1#sh ip ospf
Routing Process "ospf 1" with ID 10.128.254.1
Supports only single TOS(TOS0) routes
Supports opaque LSA
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
External flood list length 0
  Area BACKBONE(0)
    Number of interfaces in this area is 8
    Area has no authentication
    SPF algorithm executed 10 times
    Area ranges are
    Number of LSA 8. Checksum Sum 0x046606
    Number of opaque link LSA 0. Checksum Sum 0x000000
    Number of DCbitless LSA 0
    Number of indication LSA 0
    Number of DoNotAge LSA 0
    Flood list length 0

msk-donskaya-babdullakhi-gw-1#
```

Рис. 10: Проверка процесса OSPF на маршрутизаторе msk-donskaya-gw-1

Проверка OSPF на msk-donskaya-gw-1

```
msk-donskaya-babdullakhi-gw-1#sh ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.128.254.2      1    FULL/BDR        00:00:36    10.128.255.2    FastEthernet0/1.5
10.128.254.4      1    FULL/BDR        00:00:32    10.128.255.6    FastEthernet0/1.6
msk-donskaya-babdullakhi-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 24 subnets, 4 masks
C    10.128.0.0/24 is directly connected, FastEthernet0/0.3
L    10.128.0.1/32 is directly connected, FastEthernet0/0.3
C    10.128.1.0/24 is directly connected, FastEthernet0/0.2
L    10.128.1.1/32 is directly connected, FastEthernet0/0.2
C    10.128.3.0/24 is directly connected, FastEthernet0/0.101
L    10.128.3.1/32 is directly connected, FastEthernet0/0.101
C    10.128.4.0/24 is directly connected, FastEthernet0/0.102
L    10.128.4.1/32 is directly connected, FastEthernet0/0.102
C    10.128.5.0/24 is directly connected, FastEthernet0/0.103
L    10.128.5.1/32 is directly connected, FastEthernet0/0.103
C    10.128.6.0/24 is directly connected, FastEthernet0/0.104
L    10.128.6.1/32 is directly connected, FastEthernet0/0.104
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.1/32 is directly connected, FastEthernet0/1.5
C    10.128.255.4/30 is directly connected, FastEthernet0/1.6
L    10.128.255.5/32 is directly connected, FastEthernet0/1.6
O    10.128.255.8/30 [110/2] via 10.128.255.2, 00:00:54, FastEthernet0/1.5
      [110/2] via 10.128.255.6, 00:00:54, FastEthernet0/1.6
S    10.129.0.0/16 [1/0] via 10.128.255.2
O    10.129.0.0/24 [110/2] via 10.128.255.2, 00:06:29, FastEthernet0/1.5
O    10.129.1.0/24 [110/2] via 10.128.255.2, 00:05:09, FastEthernet0/1.5
O    10.129.128.0/24 [110/3] via 10.128.255.2, 00:04:49, FastEthernet0/1.5
S    10.130.0.0/16 [1/0] via 10.128.255.6
O    10.130.0.0/24 [110/2] via 10.128.255.6, 00:04:13, FastEthernet0/1.6
O    10.130.1.0/24 [110/2] via 10.128.255.6, 00:04:13, FastEthernet0/1.6
      198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.51.100.0/28 is directly connected, FastEthernet0/1.4
L    198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1
```

msk-donskaya-babdullakhi-gw-1#

Проверка OSPF на msk-q42-gw-1

```
msk-q42-babdullakhi-gw-1#sh ip ospf nei
Neighbor ID      Pri   State           Dead Time   Address        Interface
10.128.254.4      1    FULL/BDR        00:00:31    10.128.255.10  FastEthernet0/1.7
10.128.254.3      1    FULL/BDR        00:00:31    10.129.1.2     FastEthernet1/0.202
10.128.254.1      1    FULL/DR         00:00:32    10.128.255.1   FastEthernet0/1.5
msk-q42-babdullakhi-gw-1#
msk-q42-babdullakhi-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 19 subnets, 4 masks
O       10.128.0.0/24 [110/2] via 10.128.255.1, 00:06:53, FastEthernet0/1.5
O       10.128.1.0/24 [110/2] via 10.128.255.1, 00:06:53, FastEthernet0/1.5
O       10.128.3.0/24 [110/2] via 10.128.255.1, 00:06:53, FastEthernet0/1.5
O       10.128.4.0/24 [110/2] via 10.128.255.1, 00:06:53, FastEthernet0/1.5
O       10.128.5.0/24 [110/2] via 10.128.255.1, 00:06:53, FastEthernet0/1.5
O       10.128.6.0/24 [110/2] via 10.128.255.1, 00:06:53, FastEthernet0/1.5
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
O       10.128.255.4/30 [110/2] via 10.128.255.1, 00:01:23, FastEthernet0/1.5
        [110/2] via 10.128.255.10, 00:01:23, FastEthernet0/1.7
C       10.128.255.8/30 is directly connected, FastEthernet0/1.7
L       10.128.255.9/32 is directly connected, FastEthernet0/1.7
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
C       10.129.1.0/24 is directly connected, FastEthernet1/0.202
L       10.129.1.1/32 is directly connected, FastEthernet1/0.202
S       10.129.128.0/17 [1/0] via 10.129.1.2
O       10.129.128.0/24 [110/2] via 10.129.1.2, 00:05:18, FastEthernet1/0.202
O       10.130.0.0/24 [110/2] via 10.128.255.10, 00:01:23, FastEthernet0/1.7
O       10.130.1.0/24 [110/2] via 10.128.255.10, 00:01:23, FastEthernet0/1.7
S*    0.0.0.0/0 [1/0] via 10.128.255.1
```


Проверка OSPF на msk-hostel-gw-1

```
msk-hostel-babdullakhi-gw-1#sh ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address      Interface
10.128.254.2      1    FULL/DR         00:00:31    10.129.1.1   Vlan202
msk-hostel-babdullakhi-gw-1#
msk-hostel-babdullakhi-gw-1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 14 subnets, 2 masks
O       10.128.0.0/24 [110/3] via 10.129.1.1, 00:05:42, Vlan202
O       10.128.1.0/24 [110/3] via 10.129.1.1, 00:05:42, Vlan202
O       10.128.3.0/24 [110/3] via 10.129.1.1, 00:05:42, Vlan202
O       10.128.4.0/24 [110/3] via 10.129.1.1, 00:05:42, Vlan202
O       10.128.5.0/24 [110/3] via 10.129.1.1, 00:05:42, Vlan202
O       10.128.6.0/24 [110/3] via 10.129.1.1, 00:05:42, Vlan202
O       10.128.255.0/30 [110/2] via 10.129.1.1, 00:05:42, Vlan202
O       10.128.255.4/30 [110/3] via 10.129.1.1, 00:05:01, Vlan202
O       10.128.255.8/30 [110/2] via 10.129.1.1, 00:01:47, Vlan202
O       10.129.0.0/24 [110/2] via 10.129.1.1, 00:05:42, Vlan202
C       10.129.1.0/24 is directly connected, Vlan202
C       10.129.128.0/24 is directly connected, Vlan301
O       10.130.0.0/24 [110/3] via 10.129.1.1, 00:01:47, Vlan202
O       10.130.1.0/24 [110/3] via 10.129.1.1, 00:01:47, Vlan202
S*     0.0.0.0/0 [1/0] via 10.129.1.1
msk-hostel-babdullakhi-gw-1#
```

Проверка OSPF на sch-sochi-gw-1

```
sch-sochi-babdullakhi-gw-1#sh ip ospf nei

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.128.254.2      1    FULL/DR         00:00:38    10.128.255.9   FastEthernet0/0.7
10.128.254.1      1    FULL/DR         00:00:30    10.128.255.5   FastEthernet0/0.6
sch-sochi-babdullakhi-gw-1#
sch-sochi-babdullakhi-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 18 subnets, 3 masks
O       10.128.0.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.1.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.3.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.4.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.5.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.6.0/24 [110/2] via 10.128.255.5, 00:05:29, FastEthernet0/0.6
O       10.128.255.0/30 [110/2] via 10.128.255.5, 00:02:15, FastEthernet0/0.6
        [110/2] via 10.128.255.9, 00:02:15, FastEthernet0/0.7
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.128.255.8/30 is directly connected, FastEthernet0/0.7
L       10.128.255.10/32 is directly connected, FastEthernet0/0.7
O       10.129.0.0/24 [110/2] via 10.128.255.9, 00:02:15, FastEthernet0/0.7
O       10.129.1.0/24 [110/2] via 10.128.255.9, 00:02:15, FastEthernet0/0.7
O       10.129.128.0/24 [110/3] via 10.128.255.9, 00:02:15, FastEthernet0/0.7
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*    0.0.0.0/0 [1/0] via 10.128.255.5

sch-sochi-babdullakhi-gw-1#
```

Симуляция прохождения ICMP-пакета

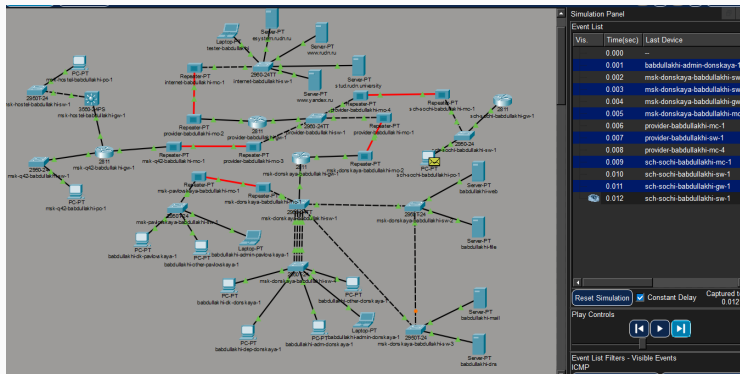


Рис. 15: Проверка прохождения ICMP-пакета в режиме Simulation

```
provider-babdullakhi-sw-1#
provider-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-babdullakhi-sw-1(config)#no int vlan 6
provider-babdullakhi-sw-1(config)#
%LINK-5-CHANGED: Interface Vlan6, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to down

provider-babdullakhi-sw-1(config)#no vlan 6
provider-babdullakhi-sw-1(config)#exit
provider-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
provider-babdullakhi-sw-1#
```

Рис. 16: Отключение VLAN 6 на коммутаторе provider-sw-1

```
C:\>  
C:\>ping 10.130.0.200  
  
Pinging 10.130.0.200 with 32 bytes of data:  
  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
  
Ping statistics for 10.130.0.200:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>
```

Рис. 17: Проверка доступности узла 10.130.0.200 после отключения VLAN 6

```
provider-babdullakhi-sw-1>en
Password:
provider-babdullakhi-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-babdullakhi-sw-1(config)#
provider-babdullakhi-sw-1(config)#vlan 6
provider-babdullakhi-sw-1(config-vlan)#name sochi
provider-babdullakhi-sw-1(config-vlan)#int vlan 6
provider-babdullakhi-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-babdullakhi-sw-1(config-if)#no sh
provider-babdullakhi-sw-1(config-if)#exit
provider-babdullakhi-sw-1(config)#exit
provider-babdullakhi-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-babdullakhi-sw-1#wr mem
Building configuration...
[OK]
provider-babdullakhi-sw-1#
```

Рис. 18: Восстановление VLAN 6 на коммутаторе provider-sw-1

Повторная проверка ICMP в Simulation

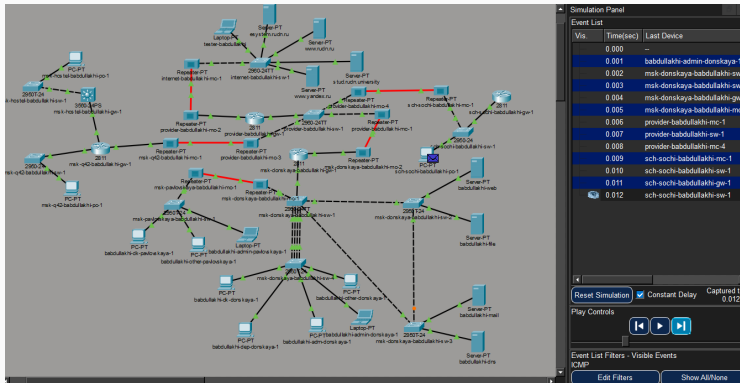


Рис. 19: Проверка прохождения ICMP-пакета после восстановления VLAN 6

Выводы по проделанной работе

В ходе лабораторной работы была настроена динамическая маршрутизация по протоколу **OSPF** между территориями организации. На маршрутизаторах был запущен процесс **OSPF 1**, назначены идентификаторы **router-id**, объявлены сети в области **area 0** и сформированы соседства OSPF.

Также была настроена прямая связь между сетью квартала 42 в Москве и филиалом в Сочи через VLAN 7 и подинтерфейсы с инкапсуляцией **802.1Q**. Проверка таблиц маршрутизации показала наличие маршрутов с признаком **O**, полученных по OSPF.

В режиме симуляции было отслежено прохождение ICMP-пакетов от ноутбука администратора сети на Донской до компьютера пользователя в филиале Сочи. После отключения VLAN 6 основной маршрут стал недоступен, а после восстановления VLAN 6 прохождение пакетов по основному каналу было восстановлено.